# Report on Attending the ATRIUM Computer Vision in Archaeology Training School in Brno

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## Introduction

From 13 to 18 September 2026, I was in Brno, Czech Republic, to attend the Computer Vision in Archaeology Training School. I arrived a day earlier, so I had the opportunity to go sightseeing. The five-day course was held by the Archaeological Information System of the Czech Republic at the Institute of Archaeology of the Czech Academy of Sciences, Brno, organised by the ATRIUM and MAIA COST Action. It brought researchers from archaeology to explore how computer vision and machine learning can detect, classify and analyse archaeological feature. The programme combined introductory lectures with practical exercises. It started with the Python basics, through image processing, annotation, model training, evaluation to the interpretation of model outputs.

My main motivation for attending was connected to two areas of my current work. I am a PhD student at the ELTE Doctoral School of History, specialising in archaeology. My dissertation is about the building history of the Inner City Parish Church in Pest, with the focus on the stone carvings. Identifying individual carvings and their stylistic relationships can be difficult, when surfaces are damaged or fragmentary. I am developing a convolutional neural network (CNN) based approach that could support identification by suggesting possible parallels for expert examination. I also work at the Institute of Archaeology, Research Centre for the Humanities, ELTE, Where I use CNN-based methods in archaeological remote sensing. As most of my programming and machine learning knowledge has been learnt through independent way, I hoped to consolidate my technical foundations and learn more systematic and reproducible ways of working with archaeological image data.

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*Figure 1 The beautiful cathedral of Brno*

# Day 1 (September 14)

The firs day introduced the main tasks of computer vision and the ways in which they can answer archaeological questions. We discussed image classification, object detection and segmentation, as well as the difference between recognising visual patterns and assigning them a meaningful archaeological interpretation. A practical introduction to zero-shot image classification with Clip illustrated how a pre-trained model can compare images with textual descriptions without first being trained on a dedicated archaeological dataset. At the same time, the exercise highlighted an important limitation, results depend strongly on the wording of the labels and on how well the model's general training data represent the archaeological object under study.

The other sessions built a common technical starting point. We worked with Python in Google Colab and used libraries such as NumPY, Matplotlib and OpenCV for basic image manipulation and visualisation. The exercises covered simple operations such as loading images, inspecting pixel values, separating an object from its background and obtaining

measurements from an image. We were also introduced to AI assisted coding and the use of coding agents.

## Day 2 (September 15)

Tuesday focused on archaeological artifact photographs and the creation of datasets suitable for machine learning. Before training a model, the researcher has to decide what the relevant classes are, how objects should be labelled and which properties must be recorded. During practical sessions we explored annotation workflows with CVAT and learnt about common data formats, including COCO and YOLO. Tools for assisted segmentation, such as SAM 2, demonstrated how suggested object outlines may speed up the annotation process while still requiring manual checking.

The afternoon included short presentations and discussions of participants' research projects. I had the opportunity to present the CNN-based system I am developing to support the identification of stone carvings. The discussion provided useful suggestions for further developing and improving the system. I also found the other participants' presentations inspiring, as they introduced different ways in which computer vision could be incorporated into archaeological research and gave me new ideas for my own methodology.

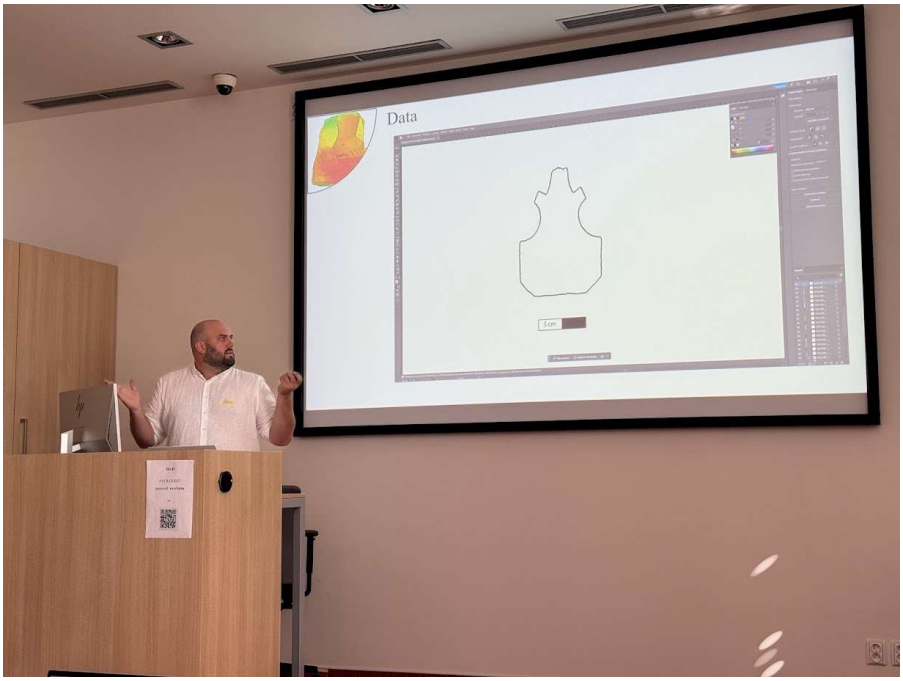



*Figure 2 Me as presented my CNN*

# Day 3 (September 16)

The third day was devoted to microscopy and the preparation of images for computer-assisted analysis. Using examples from lithic use-wear studies. We examined the challenges of working with images produced under different acquisition conditions. Variations in magnification, illumination, contrast, sharpness and colour can affect both conventional image-processing methods and machine-learning models. Consequently, an apparent visual difference between two images does not necessarily reflect a meaningful difference in the archaeological material itself.

The hands-on exercises introduced several approaches to image standardisation and feature extraction. We compared image histograms, experimented with histogram matching and contrast adjustment and explored thresholding, filtering and segmentation methods to isolate relevant areas. We also discussed how augmentation, for example rotations or changes in scale, can increase the variety of a limited training set. However, preprocessing must be chosen carefully, because aggressive enhancement or transformations may remove the very surface details that the researcher want to study. It is essential to keep the original images, document the transformations and assess whether the resulting masks or measurements correspond to the archaeological features of interest.

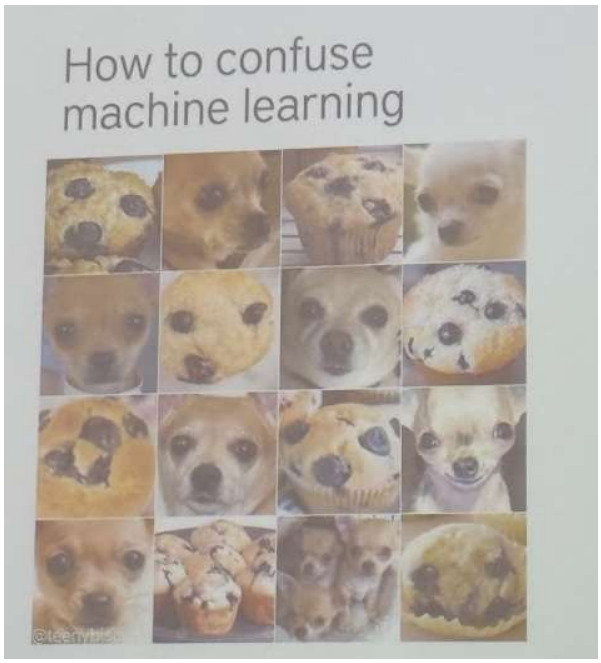



*Figure 3 My favourite slide of the week :)*

## Day 4 (September 17)

In the morning session we dealt wit remote sensing. We explored the use of Sentinel-2 multispectral satellite data to identify possible archaeological cropmarks, including a case study of the large Bronze Age fortified site at Cornești-Iarcuri. The workflow combined information from spectral bands and vegetation indices, particularly the Normalised Difference Vegetation Index (NDVI), with methods for enhancing local contrasts and linear features. The aim was to produce candidate archaeological features rather than to assume that every vegetation anomaly has an archaeological explanation.

In the afternoon, the scale of analysis changed from landscapes to small features in rock-art documentation. The case study concerned the detection and segmentation of cupmarks, using rock-art images and visualisations derived from 3D recording. We examined annotation with Label Studio, the use of a YOLO segmentation model and the advantages of processing high-resolution images in smaller overlapping tiles, for example through SAHI. Tiling is important because very small objects can disappear when an entire large image is resized to the input dimensions expected by a model. The challenges are also relevant to carved architectural fragments: fine incisions, damaged motifs and small diagnostic details may be lost if images are downscaled too aggressively. Both sessions reinforced the need to review model predictions against the original material and its archaeological context.

## Day 5 (September 18)

On the final day, we returned to artefact-based research through a case study involving coins. This provided an opportunity to bring together several themes discussed earlier in the week, from organising image datasets to combining different recognition methods. We learnt that, in some cases, it is not necessary to use the latest or most specialised AI models, as older algorithms and methods originally developed for different purposes can still be highly effective. This highlighted the importance of choosing methods based on the specific research problem rather than treating AI as a universal solution. We also learnt that AI cannot replace researchers in the classification and interpretation of archaeological artefacts. While it can support and accelerate these processes, archaeological expertise and human judgement remain essential.

## Personal reflections

The training school was particularly valuable because it connected methods that I had previously encountered mainly through independent experimentation within a coherent research workflow. Beyond the individual software packages, I gained a clearer understanding of how image acquisition, preprocessing, annotation, data management, model selection and evaluation influence one another.

For my doctoral work on the Inner City Parish Church in Pest, I would like to apply these lessons by developing a better-structured and more consistently annotated collection of carved stone data. The longer-term aim would be an assistive system that proposes visually comparable fragments for examination, with every suggestion remaining open to archaeological and art historical verification.

For my remote sensing work, the session on multispectral imagery introduced several useful techniques that I could apply to my own datasets. The lectures, practical exercises, and discussions with researchers from different countries broadened my understanding of how computer vision can be applied to various archaeological research problems. They also highlighted that researchers working in different fields of archaeology often face similar methodological challenges.

Overall, attending the ATRIUM Computer Vision in Archaeology Training School helped me move from using individual machine learning techniques towards designing more transparent, reproducible and archaeologically meaningful workflows. I am grateful to the organisers, instructors and host institution for the practical teaching and stimulating discussions and to the ATRIUM and MAIA initiatives for making the school possible.



*Figure 4 Also the beer was amazing. (It is a cut-beer)*